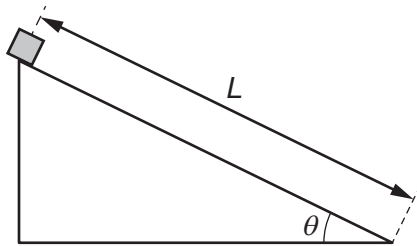


- 15** A block is released from rest at the top of a slope inclined at an angle to the horizontal. The slope has length L as shown in the diagram.



There are no resistive forces acting on the block.

What is the speed of the block at the bottom of the slope?

- A** $4.43\sqrt{L\cos\theta}$ **B** $4.43\sqrt{L\sin\theta}$ **C** $19.6L\cos\theta$ **D** $19.6L\sin\theta$

- 16** A skateboarder and her skateboard have a total mass of 70 kg . She pushes on the ground with